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Part 1 Introduction

Chapter 1 Introduction to Physical Design

Urgent need for automation { scale $\uparrow\uparrow$
some assumption became invalid
 $\downarrow$ eg.

Delay:

$$t_{\text{wire}} \propto L \rightarrow t_{\text{wire}} \propto L^2 \xrightarrow{\text{Buffer}} T_{\text{total}} \propto L$$

(寄生电容)

Interconnect Dominance

-> Physical Synthesis

-> microarchitectural design

Madonna: I'm young forever

Sub-90nm Technologies

-> Manufacturability

-> DFM

Mask Cost $\uparrow \uparrow$

-> FPGA thrive

Overview of the physical design process: synthesize, floorplanning, placement, clock tree synthesis, routing, finish